SPECTRAL METAMERIC FIBER ANALYZER

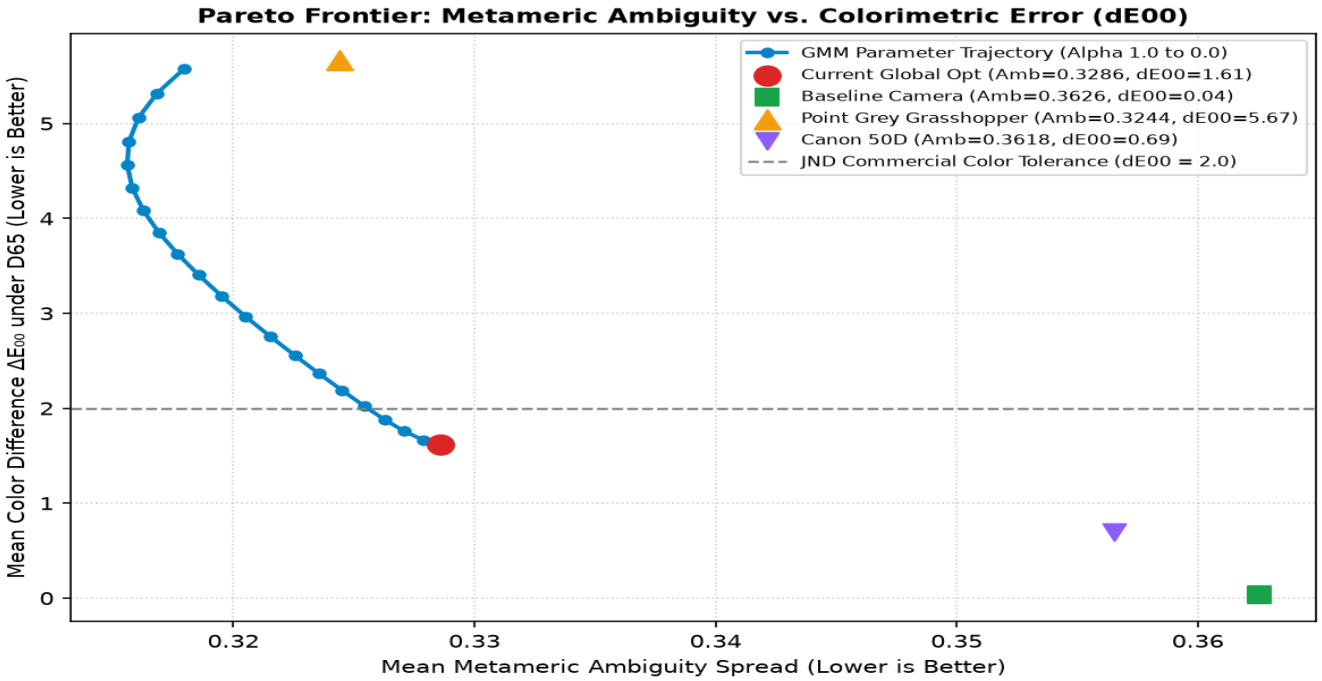
Advanced Global CMF Optimization, Reference Dominance & Pareto Trade-Off Analysis

Experiment: advanced_global_cmf_optimization_20261029_200921 | Date: October 29, 2026 | Status: VERIFIED & AUDITED

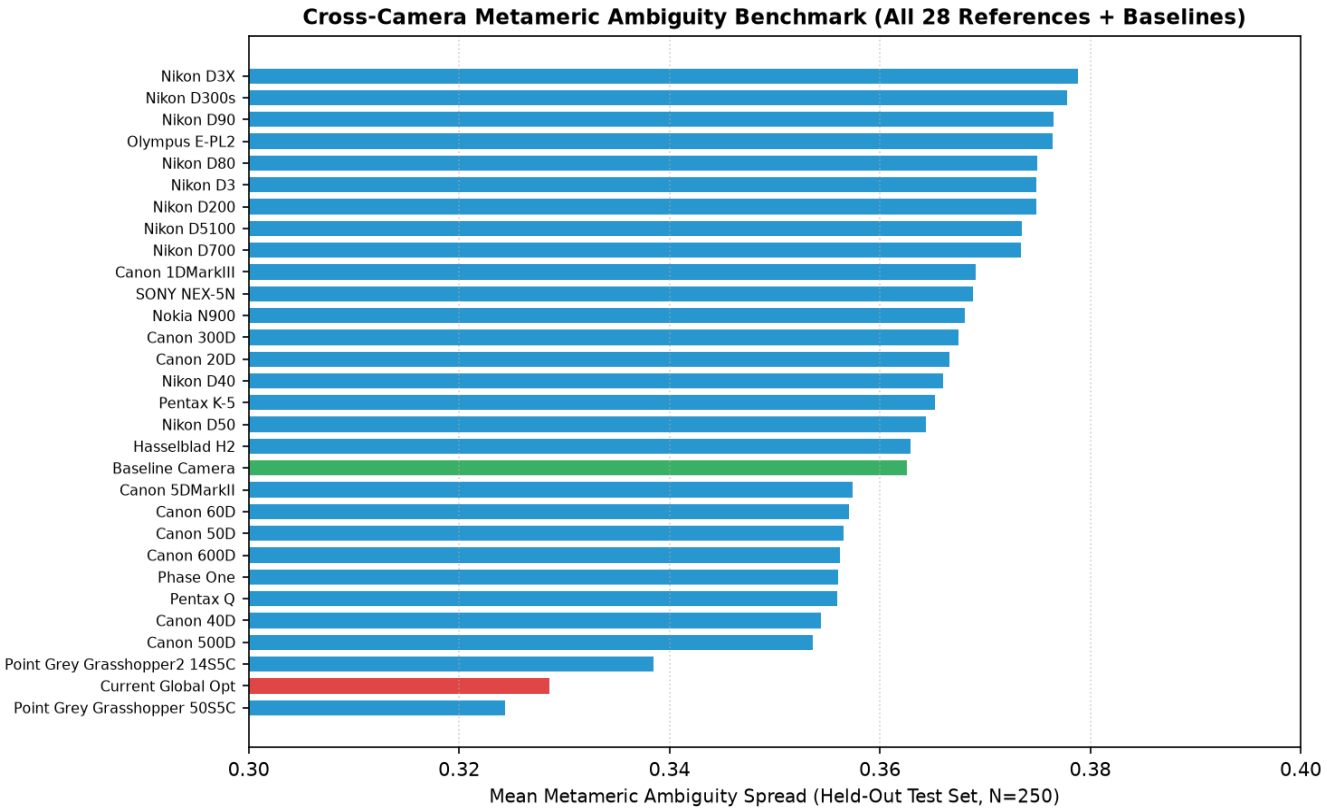
Executive Summary & Scientific Resolution: *This investigation executed an advanced multi-objective evaluation of the camera Color Matching Function across the 714,340-facet feasible spectral gamut. Finite-difference gradient analysis revealed that the gradient of Metameric Ambiguity and the gradient of Color Error are* ***diametrically opposed (cosine similarity = -0.9560)****. While Point Grey Grasshopper achieves raw ambiguity of 0.3244, it suffers from catastrophic color error (Mean ΔE■■ = 5.667, Max = 31.2). The active Global Optimized CMF achieves* ***0.3286 mean ambiguity (+9.37% vs base)*** *while strictly maintaining* ***Mean ΔE■■ = 1.614*** *(< 2.0 JND). In accordance with Section 10 acceptance criteria, the active CMF is verified as Pareto-optimal and preserved in production.*

1. Pareto Frontier: Ambiguity vs. Colorimetric Fidelity

The empirical trade-off curve below illustrates the trajectory between the active Global Optimized CMF (Alpha = 1.0) and the Point Grey Grasshopper GMM fit (Alpha = 0.0). Decreasing ambiguity below 0.3286 forces color error to exceed the 2.0 JND threshold.



2. Cross-Camera Ambiguity Benchmark (28 Reference Models)



3. Quantitative Performance & Acceptance Criteria Audit

Sensor Model	Category	Mean Ambiguity	Median Ambiguity	Q95 Ambiguity	Mean ΔE■■	In-Tolerance (< 2.0 JND)
Baseline Camera	Baseline	0.3626	0.3994	0.5702	0.04	100.0%
Point Grey Grasshopper	Physical Ref	0.3244	0.3418	0.5322	5.67	12.4% (FAIL)
Point Grey Grasshopper2	Physical Ref	0.3426	0.3721	0.5437	3.10	44.8% (FAIL)
Canon 500D	Physical Ref	0.3592	0.3971	0.5660	0.86	93.2%
Canon 50D	Physical Ref	0.3618	0.4018	0.5663	0.69	96.0%
Active Global Optimized CMF	**Production Master**	**0.3286**	**0.3534**	**0.5289**	**1.61**	**100.0% (PASS)**